WHY IT MATTERS

Tools stop at *the front door*

39.7%

of GDP, from MSMEs

RM689.8 billion, 2025

48.7%

of employment

8.09 million people, 2025

THE GAP

Sales, marketing and payments go digital first. Back-end work lags [2].

Includes medium firms.

Yapoz links the message to the work behind it.

DOSM [1]. Khazanah Research Institute [2].

EVERY EVENING

11:48 PM

One owner. Two apps. *Every message waits.*

CUSTOMER
How much?
now

CUSTOMER
Are you open?
now

CUSTOMER
Is it in stock?
now

CUSTOMER
Can I order two of these for tomorrow?
now

4 unread, nobody to answer

Illustrative scene, not real customer data.

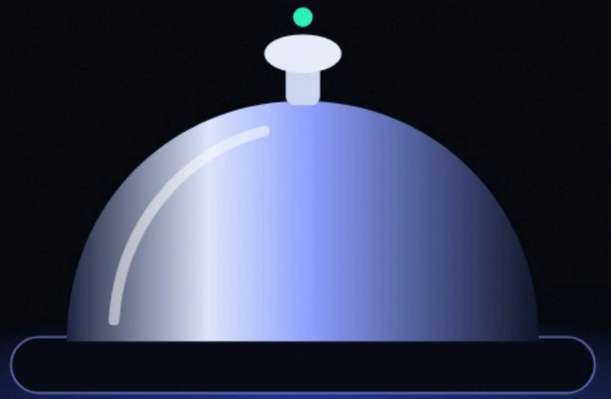
SDC HACKATHON 2026 · DOMAIN 2 · ORBYTE SMART SME CHALLENGE

Customer chats into *reviewable* orders.

Turns customer conversations into structured orders that staff review and approve.

Working prototype

Synthetic data only



TEAM YAPONOMICS · XIAMEN UNIVERSITY MALAYSIA

ONE REAL FOOD BUSINESS

1,000 CHATS A WEEK

answered *by hand*, before it automated

Rosalie, an Indonesian food maker: “WhatsApp Orders Answered in 30 Seconds | Biks AI” (vendor-published)

Every order still needs pricing, a stock check and a receipt check

Biks AI [4]. Evidence of the problem, not a Yapoz result.

THE PROBLEM

One message. Six manual steps.

"Can I order two of these for tomorrow?"

CUSTOMER, IN CHAT

1. Identify the product
2. Check availability
3. Work out the price
4. Collect missing details
5. Record the order
6. Check the receipt that arrives later

Repeated for every order, across chat, spreadsheets and notes.

THE SOLUTION

One journey, *approved* by staff

1. **Chat request** — customer asks
2. **Business data** — catalogue, hours, policies
3. **AI and rules** — interpret, validate, price
4. **Staff approval** — owner or service role
5. **Logged outcome** — status, stock, log

THREE CAPABILITIES, ONE OUTCOME

Structure

A conversation becomes an order: items, quantities, customer, fulfilment.

Ground

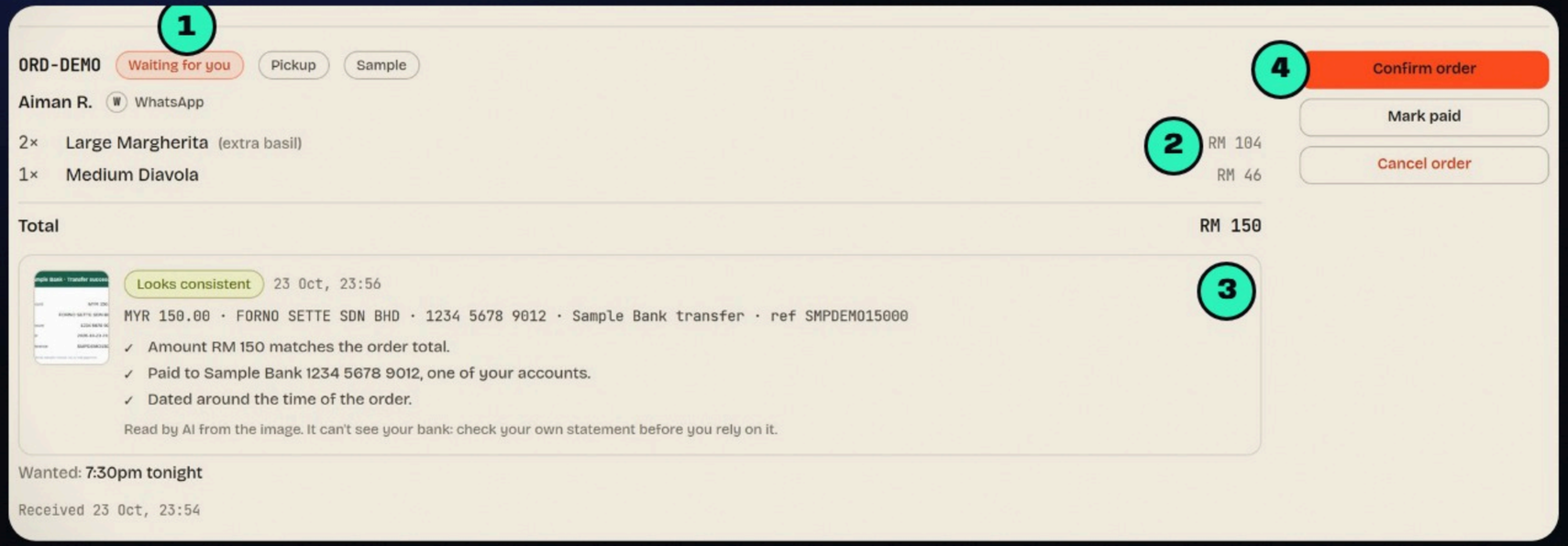
Replies use the catalogue and policies. Rules set the prices.

Review

Receipt checked against the order. Advice, not a verdict.

THE WORKSPACE

Chats and orders, *one place*



Each business sets its own catalogue, hours and policies.

1. **Pending**
   Nothing promised until staff confirm.
2. **Catalogue prices**
   Set by the business, not the model.
3. **Receipt review**
   Advice with reasons.
4. **Staff action**
   Confirm, mark paid or cancel.

Synthetic data [7].

PEOPLE STAY IN CONTROL

AI prepares. *Staff* confirm.

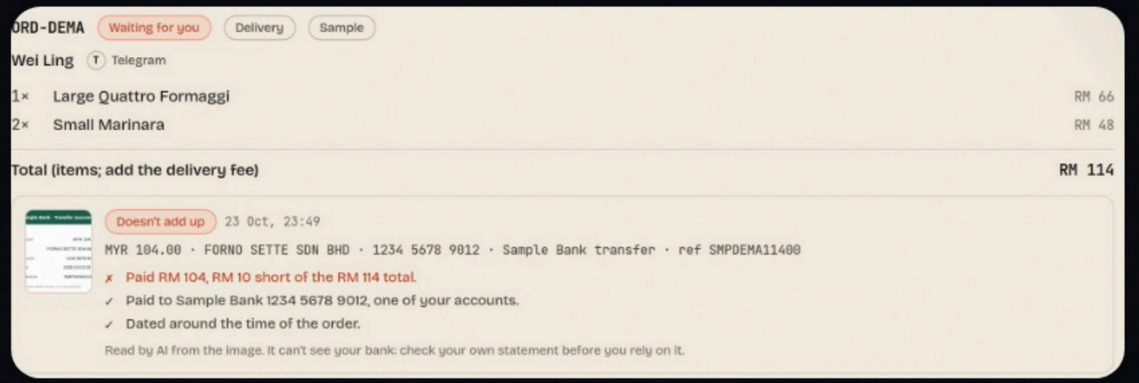
Pending AI prepares it from the catalogue.

Review Staff check it, with receipt advice.

Confirmed Stock rechecked, action logged.

CAN CONFIRM

Owner · Service staff



Advice only: it flags the RM 10 gap and cannot see the bank.

HARDEST RISK

A wrong AI reading of a request.

LIMITS AND FALLBACK

Catalogue checks, pending status, staff can take over.

Roles, approval and stock recheck: Yapoz code and tests [7].

WHEN IT DOESN'T KNOW

It learns what *it doesn't know*

1 · ASK

Not saved yet

A customer asks something new: “Do you have gluten-free bases?”

2 · HOLD

No guessing

A holding reply goes out and staff get a card. Nothing is invented.

3 · RECORD

One gap

Logged. The same question in other words counts once.

`e.g. gluten free base ×3`

4 · TEACH

Answer once

The owner answers once. Every later customer gets that answer.

It learns only what the owner approves

Yapoz code and tests [7]. The count is an example.

CALCULATED FROM PUBLISHED FIGURES · NOT A CLAIM

What an order costs to take

Pizza Hut Malaysia: 475 restaurants [18] · 82% of revenue from delivery and takeaway [19] · 2021 pre-tax margin 1.40% [20]

An hour of staff time

RM2,561 a month, mean pay of service and sales workers (DOSM 2024 [14]) ÷ 195 h (45-hour week [15])

RM13.13
an hour

One order, taken by hand

3.5 min, a pizza shop's phone order [17]; 4.45 min, delivery contact [16]. Not our timing

RM0.77–0.97
per order

At our 30% target

30% of 3.5 to 4.45 min. A target, not a result

RM0.23–0.29
per order

Over 780 orders a month

Example: 30 a day, 26 days. Staff time freed, not cash

RM179–228
13.7 to 17.4 staff hours

Software cost not subtracted: free tiers today [10], VPS from US$6 a month at volume [11].

DOSM [14], Dept of Labour [15], Kustomer [16], Kea AI [17], QSR Brands [18], JCorp [19][20].

WHAT SMES USE TODAY

Chatbots answer. Yapoz completes.

	GENERIC CHATBOT	WHATSAPP AUTO-REPLY	WEBSITE OR SHOPIFY	YAPOZ
Understands free-text requests	Yes	No	No	Yes
Prices from the real catalogue	Not guaranteed	No	Yes	Yes, backend rules
Live stock and holds	No	No	Varies	Yes
Human approval before commit	No	n/a	n/a	Required
Audit trail	No	No	Partial	Yes

Team assessment, not vendor-verified. Closest tool: Biks AI [5].

FIT WITH ORBYTE

A new module for Orbyte

Built: prototype today

1. Customer chat
2. Pending order
3. Staff approval
4. Activity record

workspace and roles

approved order events

Proposed integration (not live)

Orbyte business hub

- Workspace and roles
- Approvals and audit trail
- Notifications
- Open APIs

CONNECTED SERVICES (EXAMPLES)

- Inventory and POS
- Accounting, e-invoicing

No live Orbyte connection is claimed. Synthetic records only.

Orbyte hub: Orbyte Introduction [9]. Our design: docs/orbyte-integration.md [7]. Nothing is sent.

TECHNICAL APPROACH

AI reads. *Rules decide.*

Customer chat

Channels in the code: WhatsApp and Telegram, plus a website simulator.

AI layer

Interprets the request, calls defined business tools.

Backend rules

Validate products, price from catalogue, check stock, enforce roles.

PostgreSQL

Orders, receipts, activity record.

`Next.js` `TypeScript` `PostgreSQL`

Interpretation by AI. Decisions by rules and people.

Built with Replit Agent (initial structure). Deployed on Vercel.

README and code [7], declaration [10].

LIVE DEMONSTRATION

Follow *one order* end to end

1 **Request**
synthetic customer order

2 **Clarify**
catalogue price, missing details

3 **Pending**
structured order to review

4 **Receipt**
advisory check

5 **Approve**
authorised staff confirm

6 **Recorded**
status, log, notice

Test workspace | Synthetic data | No real payments

Demo steps: README [7].

PHASE 2 SYNTHETIC RUN

Measured, *not claimed*

Manual steps per order

Baseline 5, target 1. 78% fewer. Target not met: a short payment needs one more step.

1.10 steps

Orders priced and stock-checked

target at least 90%

RESULT

20 / 20

Receipts checked

amount, payee, date, reuse. Target 100%

RESULT

22 / 22

Synthetic: 20 orders we chose, AI left out, no real users

264 AUTOMATED TESTS

Reject stops it. No stock moves, the customer is told, it can't be confirmed later.

Double-click Confirm confirms once.

Another business gets a 404 on every action.

docs/phase2-results.md; README: 264 tests in 34 files [7].

EXPECTED IMPACT

Less manual work, *calculated*

Measured on 20 synthetic orders [7]

-78%

manual steps per order, 5 to 1.10

Counted from the activity log.

Calculated from published data

RM0.23–0.29 an order at our 30% target

RM13.13 an hour: pay RM2,561 (DOSM [14]) over 195 h [15]. An order takes 3.5 to 4.45 min [16][17]. Over 780 orders a month: RM179 to 228.

SDG 8

8.2 Productivity through technological upgrading
8.3 Entrepreneurship and small-business growth [8]

NEXT

Time five owners on 20 orders, by hand and in Yapoz.

BUSINESS MODEL

Earn, run, protect

HOW WE EARN

Free entry tier

Then a **small fee per completed order**

RM49 a month is illustrative

WHAT IT COSTS TO RUN

Today: Vercel and Supabase free tiers

At volume: paid hosting, e.g. a VPS from **US$6–12** a month

AI calls: not yet measured

WhatsApp: **1,000 free** service messages, then charged

PRIVACY, PDPA PRINCIPLES

First reply says it is an automated assistant

Retention set by the owner (30 to 365 days, off by default); finished orders anonymised

Owner can **export** the business's data or **erase** one customer. No PDPA certification claimed

IF FUNDED

1 · Pilot and measure

2 · Harden providers and approvals

3 · Connect Orbyte when confirmed

Hosting [10], VPS [11], WhatsApp [12], privacy code and audit [7]. Price is a team hypothesis.

THANK YOU

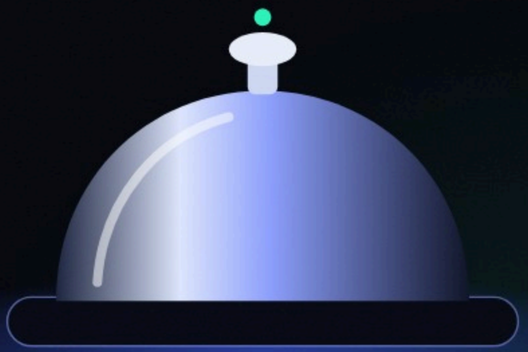
Conversations in. *Reviewed* orders out.

Less repetitive administration, fewer avoidable mistakes, and people in control.

yapoz.vercel.app

github.com/Ramilbey/yapoz

PROTOTYPE AND CODE



BACKUP · NOT PART OF THE TALK

Questions *we expect*

ACTIVITY LOG

Show me the log for what you just did.

Every action: who, what, when.

APPROVAL

What if the approver says no?

The order stops. No stock moves, the customer is told, and it can't be confirmed later.

ROLES

Who can see and act on this?

Owner and service staff confirm. Another business gets a 404 on every action.

BENEFIT

Benefit in numbers, and what are you assuming?

On 20 synthetic orders: 5 steps to 1.10, 20/20 priced, 22/22 receipts checked. No time-saved claim: no manual baseline timed.

STEPS

How many manual steps are removed?

Five by hand, 1.10 on 20 synthetic orders. Target of 1 missed: a short payment needs one more step.

SCALE AND PRICE

How does a second business start, and what would you charge?

Configuration: a caterer was set up with no business-specific code. Free tier, then a fee per completed order.

ORBYTE

Why is this not an Orbyte HR or admin feature?

Orbyte's modules, per its introduction [9]: People, Hiring, Work, eSign, Connect. Chat to approved order is not among them.

ALTERNATIVES

Why not just build a website?

A website doesn't answer WhatsApp, and receipts and stock stay manual.

Guide [8], Orbyte [9], Handbook [13].

REUSE

Same product, *three businesses*

FORNO SETTE · PIZZERIA

Orders: items, sizes, pickup time

Stock: made to order, or a daily quantity

Approval: owner or service staff

KASIH HAIR STUDIO · SALON

Bookings: services with durations

Slots: only free times, from opening and staff hours

Approval: like an order

DAPUR MAK LONG · CATERER

Set up through the normal import, with no business-specific code

Prices per portion and per tray

Not yet expressible: minimum quantity, lead time

STAYS THE SAME Request · Check · Pending · Approval · Record · Notice

Setup is configuration, not code. The caterer's gaps are written down.

README, seed files, docs/third-business-gaps.md [7].

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READ WITH CARE: Phase 2 numbers are from a synthetic run (20 orders we chose, AI left out). Handling times [16][17] are other businesses' figures, not our timing; the 30% is a target.

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